

DEEP LEARNING & COMPUTER VISION

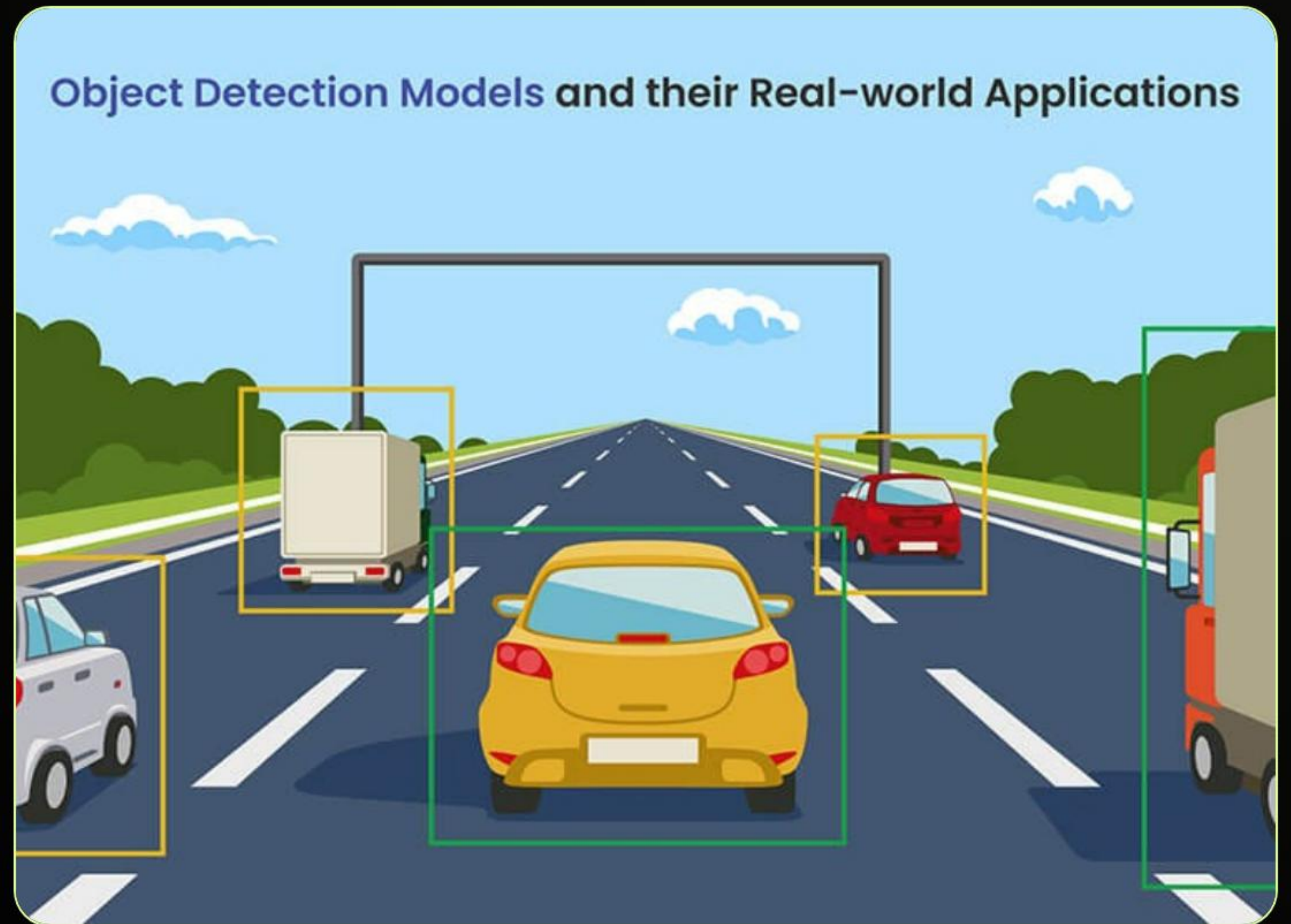
GTSRB Traffic Sign Recognition using CNN

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Project Introduction

Traffic Sign Recognition (TSR) is a critical component for the safety of autonomous vehicles and driver assistance systems.

- 🛡️ Ensuring road safety through real-time detection.
- 🚗 Fundamental pillar for Level 5 autonomy.
- 🧠 Human-level perception in complex urban environments.



Dataset Overview



GTSRB Benchmark

Official German Traffic Sign Recognition Benchmark dataset.



43 Classes

Categorized signs including speed limits, warnings, and prohibitions.



39,000+ Images

Large-scale dataset for training robust neural networks.

Data Preprocessing

Image Normalization

Resizing images to a fixed 32×32 resolution and scaling pixel intensities to the [0, 1] range for faster convergence.

Efficient tf.data Pipeline

Implementation of a robust pipeline with caching and prefetching to maximize GPU utilization during training.

Data Augmentation

To prevent overfitting and improve generalization, real-time spatial transformations are applied to the training set.

- ↻ Random Rotations
- ✚ Spatial Translations
- 🔍 Random Zooming



CNN Architecture (1/2)



3-Block Structure

Hierarchical feature extraction using deep convolutional blocks.



Batch Norm

Applied after each block to stabilize the internal covariate shift.



MaxPooling

Downsampling layers to reduce spatial dimensions and focus on features.

CNN Architecture (2/2)

50%

Progressive Dropout

Strategically increasing dropout rates from 25% up to 50% in the final layers to prevent neural co-dependency.

Dense Classification

Dual-layered fully connected head (512 & 256 units) for complex feature mapping before the Softmax output.

Training Strategy

- ⚡ **Adam Optimizer:** Adaptive moment estimation with a learning rate of 0.001.
 - 📈 **Loss Function:** Sparse Categorical Crossentropy for multi-class classification.
 - ✍️ **Validation Split:** 20% of data used for real-time performance tracking.
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Smart Callbacks

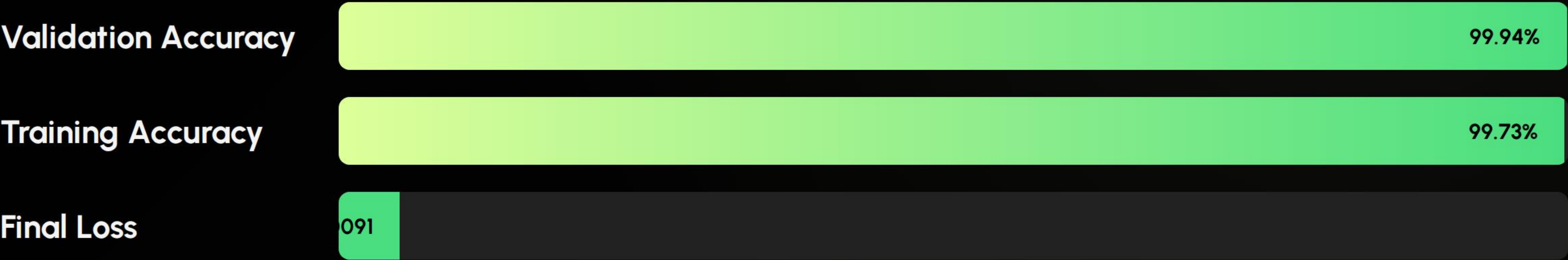
EarlyStopping

Patience set to 6 epochs. Automatically stops training when validation loss stops improving to restore the best weights.

LR Reduction

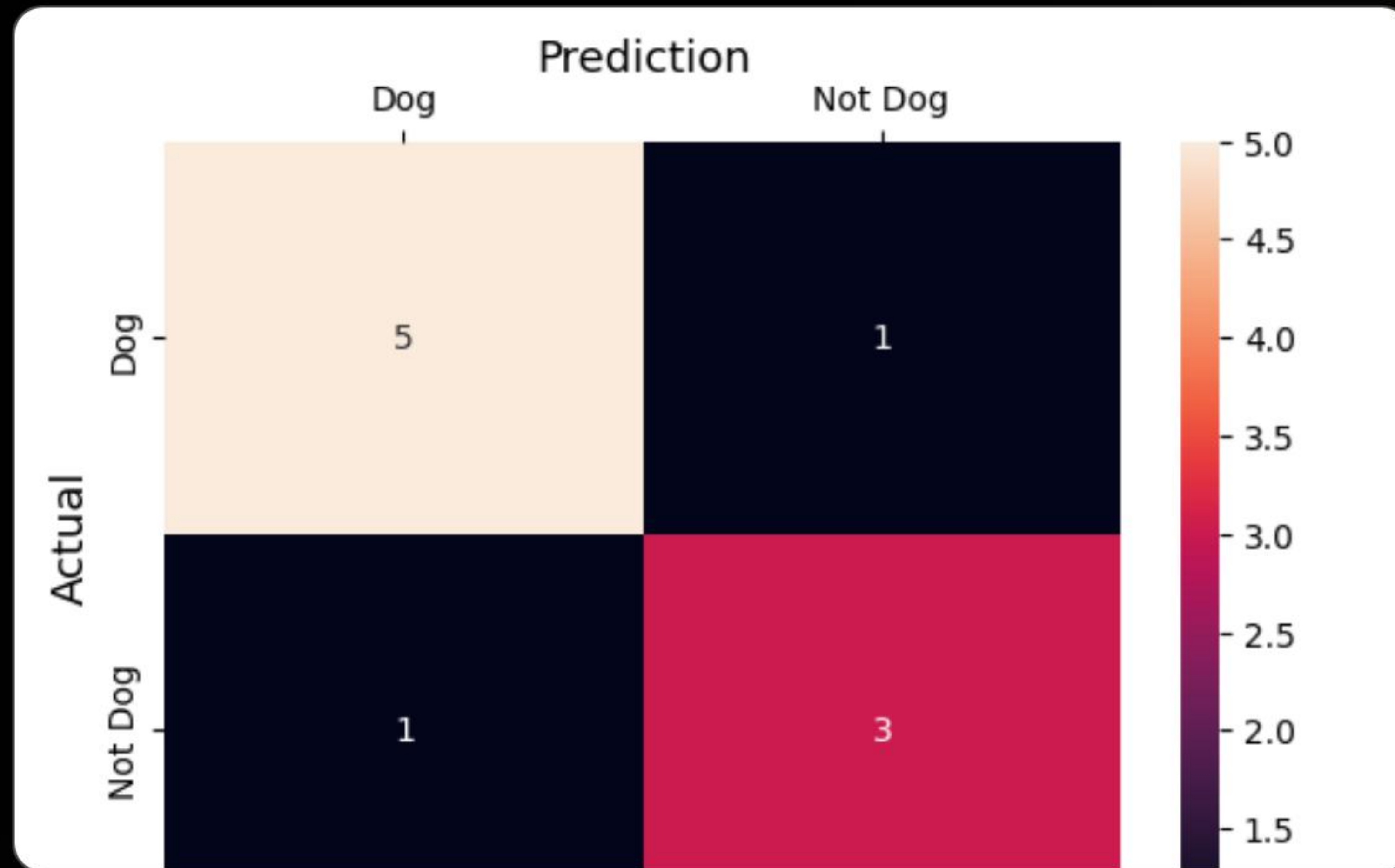
ReduceLROnPlateau halves the learning rate if the model plateaus, ensuring we reach the global minimum loss.

Final Performance Metrics

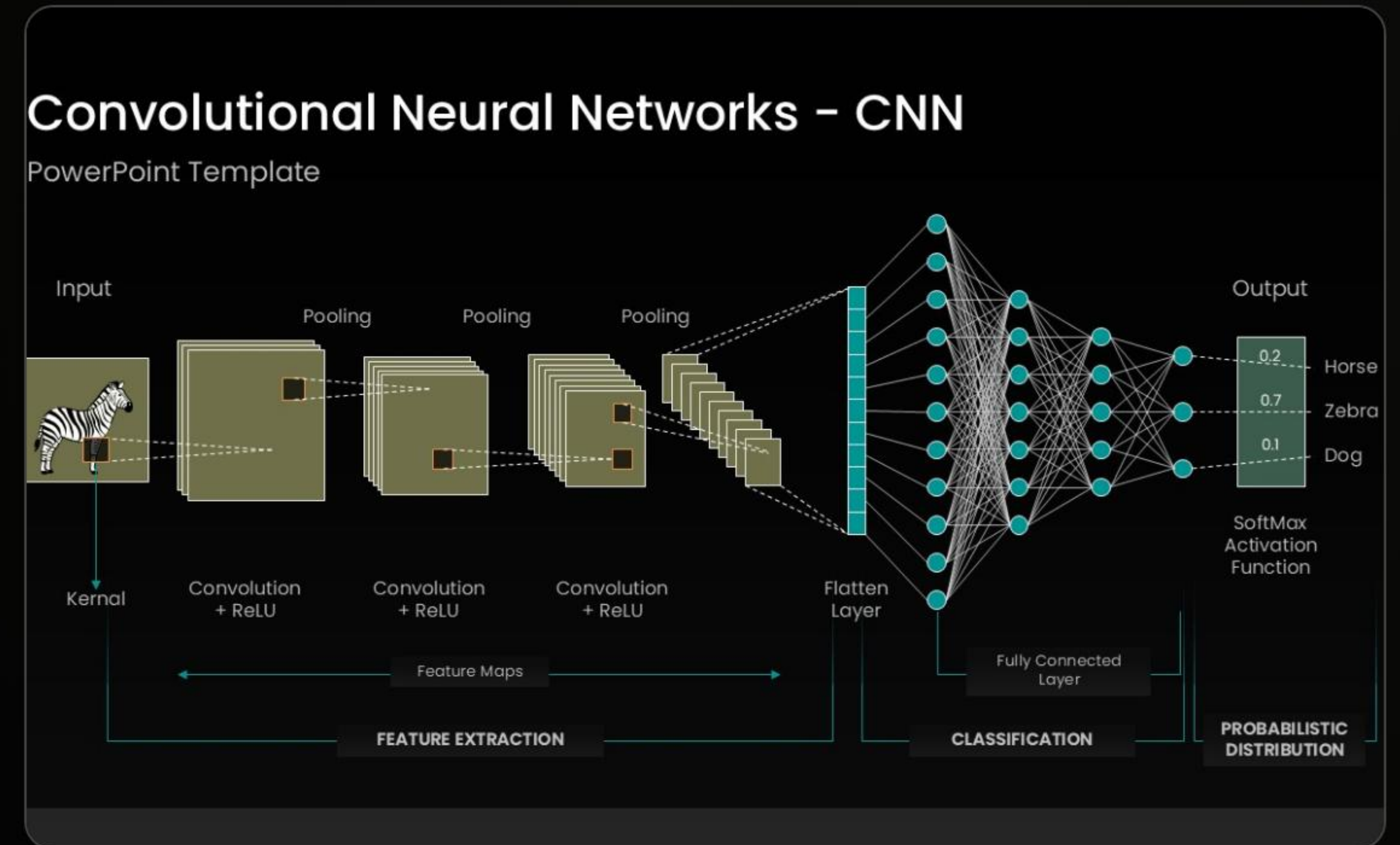


Exceptional convergence with minimal divergence between train and validation sets.

Visual Evaluation



Confusion Matrix Analysis



High-Confidence Predictions

Project Complete

The model demonstrates human-level accuracy on traffic sign classification benchmarks.

 [View Repository on GitHub](#)

Image Sources



https://miro.medium.com/1*ScMVBWolZHup8d-gnWfoHw.jpeg

Source: ai.plainenglish.io

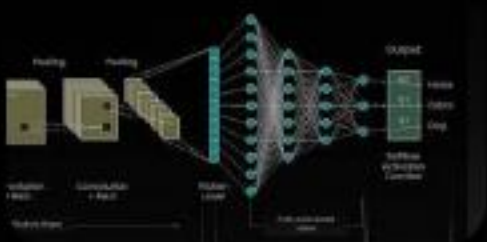


Thumbnail for https://pyimagesearch.com/wp-content/uploads/2019/11/traffic_sign_recognition_featured.jpg
Source: pyimagesearch.com



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